

OIL ANALYST

EV Sales Acceleration Poses Downside Risk to Global Oil Demand

- To assess the potential hit from the Hormuz-related oil supply shock to long-term global oil demand, we document the acceleration in Electric Vehicle (EV)¹ passenger car sales. We estimate a potential hit to global oil demand in December 2027 of 0.13-0.32mb/d (or 0.1-0.3% of global demand), depending on the persistence of the realized EV car sales acceleration ([Exhibit 5](#)).
- Our new nowcast² shows a broad-based acceleration in EV car sales as a percent of total car sales (i.e., seasonally adjusted EV penetration) since February:
 - Global EV car sales penetration has increased by 3.4pp and reached its all-time high last month of 26.1%, excluding September 2025 (when US EV sales surged in advance of a tax credit expiry) ([Exhibit 1](#)).
 - China has experienced the largest gains, with the EV penetration rate up 11.4pp ([Exhibit 2](#)).
 - Of the 15 largest EV markets, 12 have experienced an increase in EV penetration ([Exhibit 4](#)).
- We estimate the potential hit to global oil demand in December 2027 from the cumulative increase in EV sales under two scenarios³.
 - We apply our [rule-of-thumb](#) that every 1 million shift in sales from ICE to EV cars⁴ lowers road oil demand by 30kb/d in the US and 20kb/d elsewhere.
 - Under the “Temporary Acceleration” scenario, which assumes that regional EV penetration rates remain flat at their May 2026 levels, we estimate a

Alexandra Paulus
+1(212)902-7111 |
alexandra.paulus@gs.com
Goldman Sachs & Co. LLC

Daan Struyven
+1(212)357-4172 |
daan.struyven@gs.com
Goldman Sachs & Co. LLC

Yulia Zhestkova Grigsby
+1(646)446-3905 |
yulia.grigsby@gs.com
Goldman Sachs & Co. LLC

Filippo Cuscito
+44(20)7051-9073 |
filippo.cuscito@gs.com
Goldman Sachs International

¹ Electric Vehicles are defined as the sum of Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs); they exclude traditional Hybrid Vehicles (HEVs).

² We nowcast monthly EV and total (defined as the sum of EV and Internal Combustion Engine, or ICE) car sales on a seasonally adjusted basis by summing estimates for (1) “visible” countries, for which we nowcast the latest realized month of sales where missing, and (2) “invisible” countries, for which we don’t have high-frequency data. We nowcast missing months of EV and total car sales in each country using the historical relationship between monthly sales and the rolling short-term historical trend. We estimate monthly sales in “invisible” countries within a region by applying an estimated moving “coverage ratio”, defined as sales in countries for which we have high-frequency data divided by sales across the entire region.

³ For simplicity’s sake, we assume a counterfactual scenario in which regional EV car sales remain flat from June 2026 onwards, which implies some downside risk to our estimates of the boost to EV sales and the hit to oil demand.

⁴ According to our nowcast, total passenger car sales volumes have remained mostly flat since February on a seasonally adjusted basis, implying that changes in the EV car sales penetration rate have largely reflected substitution.

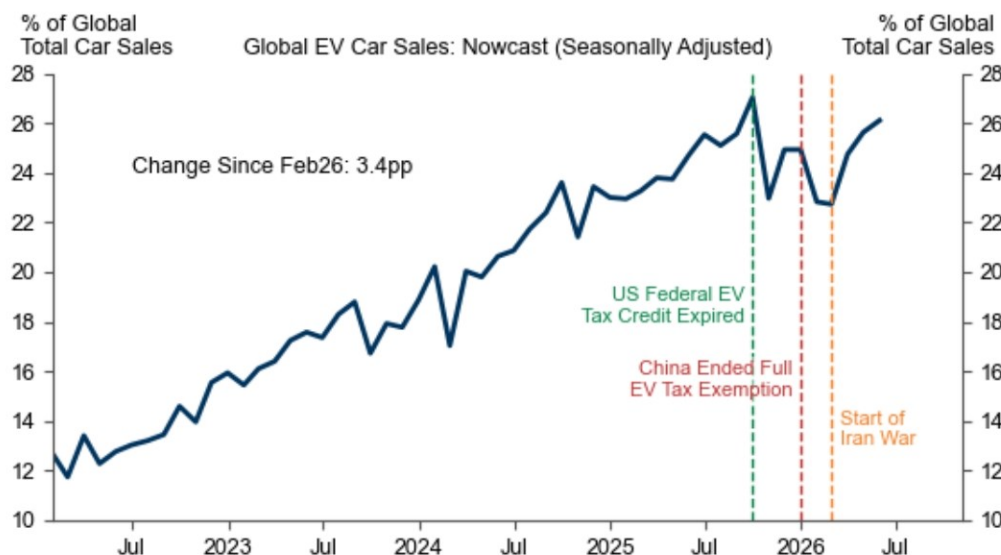
global demand hit of 0.13mb/d (Exhibit 5).

- Under the “Persistent Acceleration” scenario, which assumes that regional EV penetration rates grow linearly according to their February 2026 – May 2026 trends (through August 2026 before gradually slowing for China, and through December 2027 for world ex China), we estimate a demand hit of 0.32mb/d (Exhibit 5).
- This acceleration in global EV car sales suggests our downside oil price scenario is plausible (although potentially more persistent constraints on Hormuz flows pose significant upside price risks, too). In this downside scenario, global oil demand experiences a 1.5mb/d Hormuz-related persistent loss (vs 0.5mb/d in our base case), and Brent declines to the mid-\$50s/bbl in late 2027 (with supply beats also weighing on prices). Higher global EV passenger car sales is just one channel via which the Hormuz shock may weigh on long-term oil demand, even if oil products prices moderate further, for the following reasons:
 - **Sales vs Stock:** We are only considering *additions* to the stock of passenger cars, but substitution is already occurring *within* the existing stock in response to high fuel prices⁵, as evidenced by the just over 20% year-over-year drop in China gasoline and related products sales volumes and the surge in EV charging volumes.⁶
 - **Beyond Passenger Cars:** We omit from this analysis all non-passenger car EVs, and the around 55% of oil demand not related to road fuels (e.g., petrochemicals, for which coal might be an alternative). Most notably, two-/three-wheeler EVs comprise a majority of total EV sales in India (92% in 2025), Vietnam (80%), and China (35%) and can displace a sizable one-third to one-half of the fuel that a passenger car EV can.

⁵ For example, owners of Plug-in Hybrid EV cars may choose to charge on electricity rather than drive on gasoline, and households that own both EV and ICE cars may opt to drive their EVs more frequently.

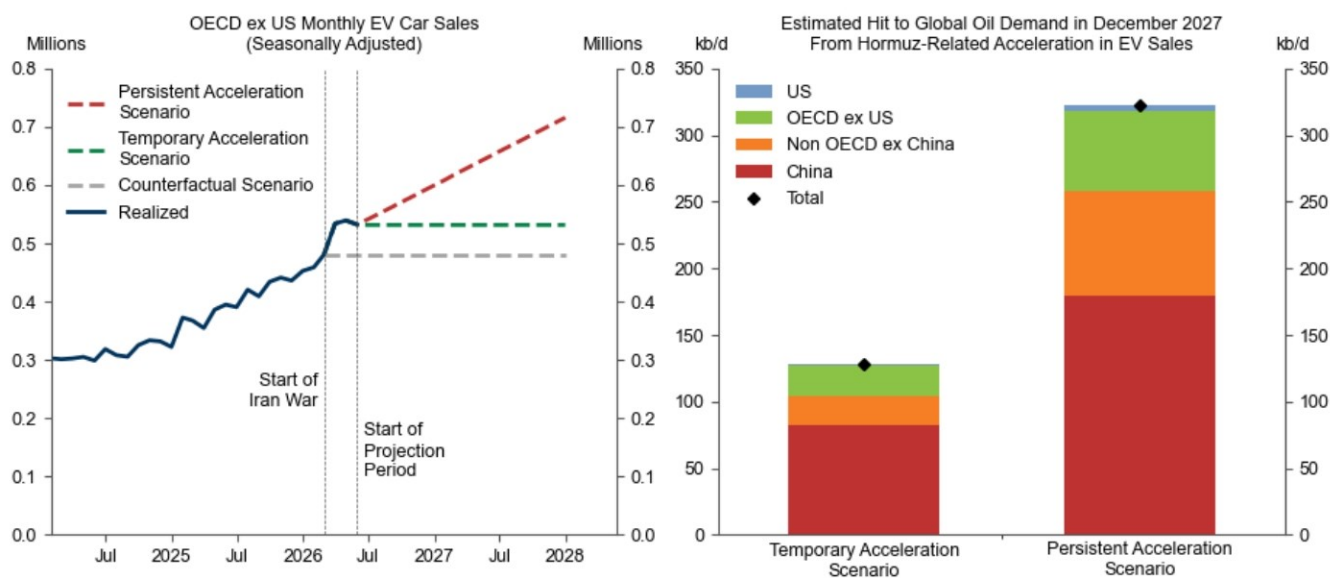
⁶ These trends in the realized data have been driven by both an increase in the total number of EVs on the road and substitution to EVs within the existing stock, but the contribution from the former has been significantly smaller. This is because the extent of turnover (=sales) that can occur within a few months, given the typical around 15-year lifespan of a passenger car, only increases the total stock by a limited amount.

According to Our Nowcast, Global EV Car Sales Penetration Has Increased By 3.4pp Since the Beginning of the Hormuz Shock



Source: Government Data, IEA, Goldman Sachs Global Investment Research

We Estimate a Hit to Global Oil Demand in December 2027 of 0.13-0.32mb/d, Depending on the Persistence of the Acceleration in EV Car Sales

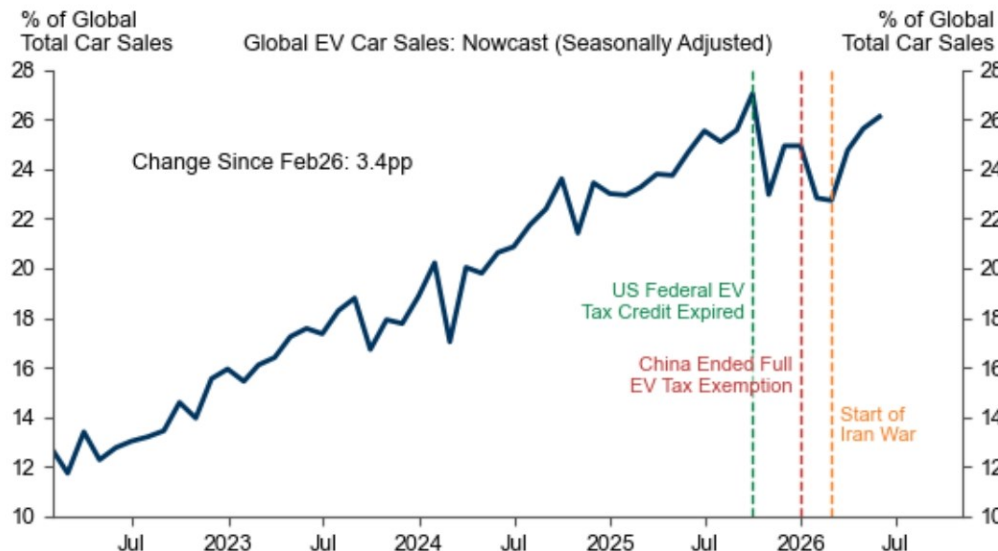


The “Temporary Acceleration” scenario assumes that regional EV car penetration rates remain flat at their May 2026 levels. The “Persistent Acceleration” scenario assumes that regional EV car penetration rates grow linearly in accordance with their respective February 2026 – May 2026 trends (through August 2026 before gradually slowing for China, and through December 2027 for world ex China). We assume in both scenarios that total passenger car sales remain flat at February 2026 – May 2026 average levels from June 2026 onwards. The hit to global oil demand is estimated relative to a counterfactual scenario in which regional EV car sales remain flat at their February 2026 levels.

Source: Goldman Sachs Global Investment Research

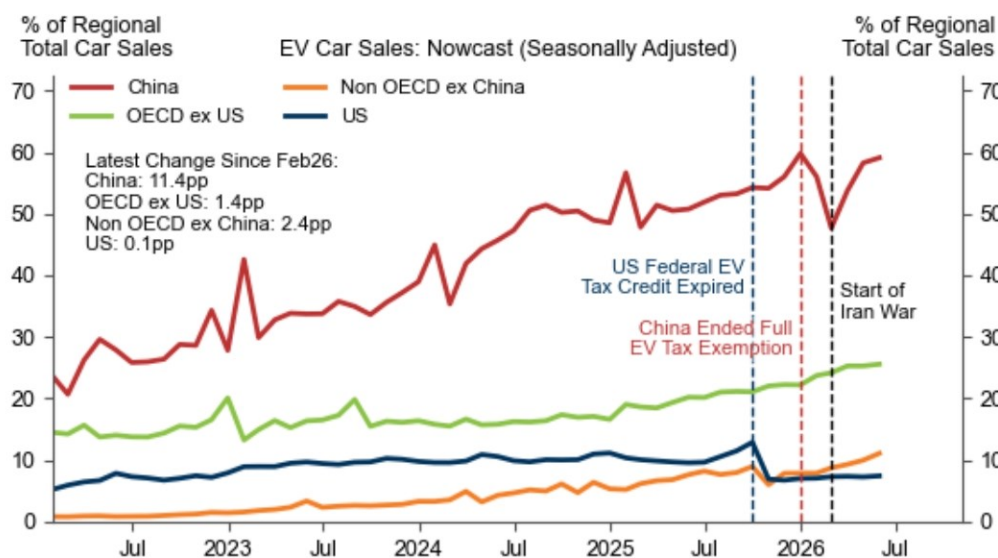
EV Sales Acceleration Poses Downside Risk to Global Oil Demand

Exhibit 1: According to Our Nowcast, Global EV Car Sales Penetration Has Increased By 3.4pp Since the Beginning of the Hormuz Shock



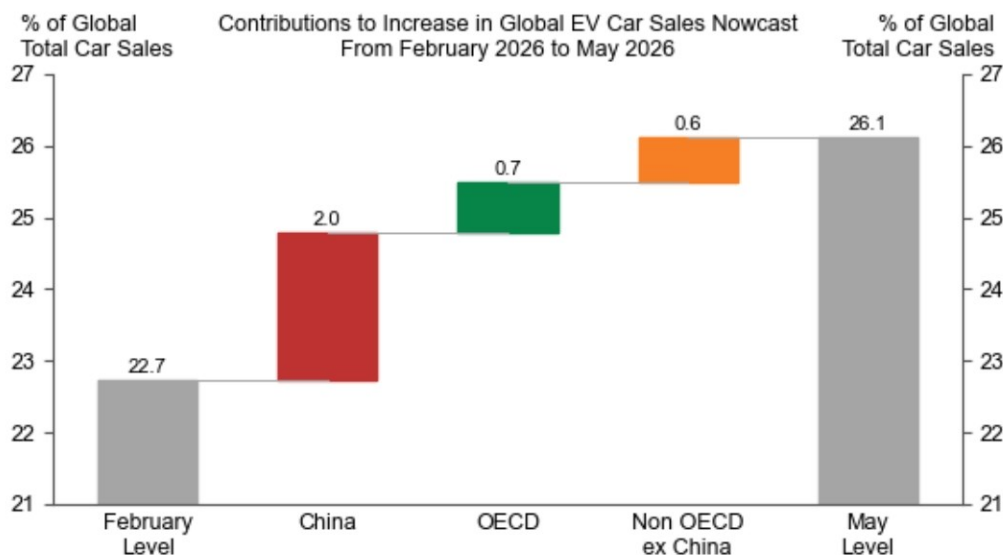
Source: Government Data, IEA, Goldman Sachs Global Investment Research

Exhibit 2: Across Regions, China Has Experienced the Largest Increase in EV Car Sales Penetration Since February of 11.4pp



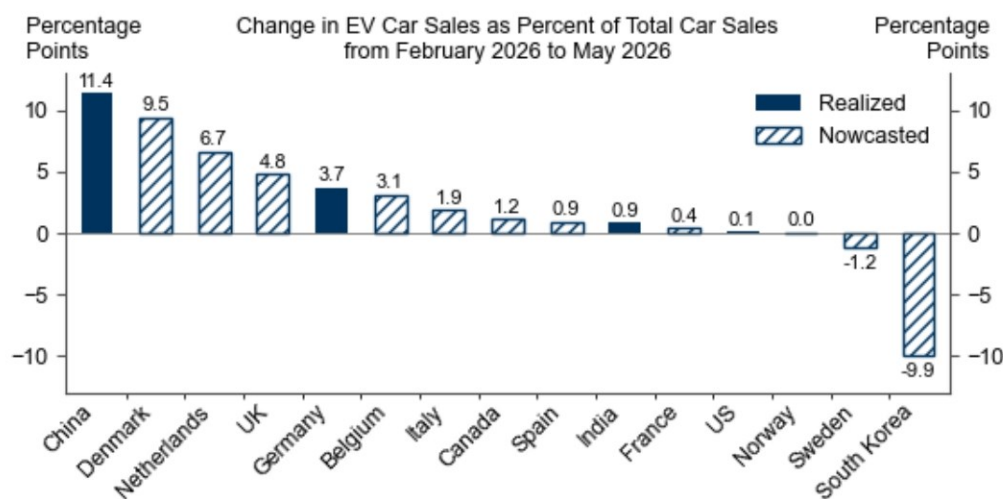
Source: Government Data, IEA, Goldman Sachs Global Investment Research

Exhibit 3: The Increase in Global EV Car Sales Penetration Since February Has Been Driven 61% By China, 21% By OECD, and 19% By Non OECD ex China



Source: Government Data, IEA, Goldman Sachs Global Investment Research

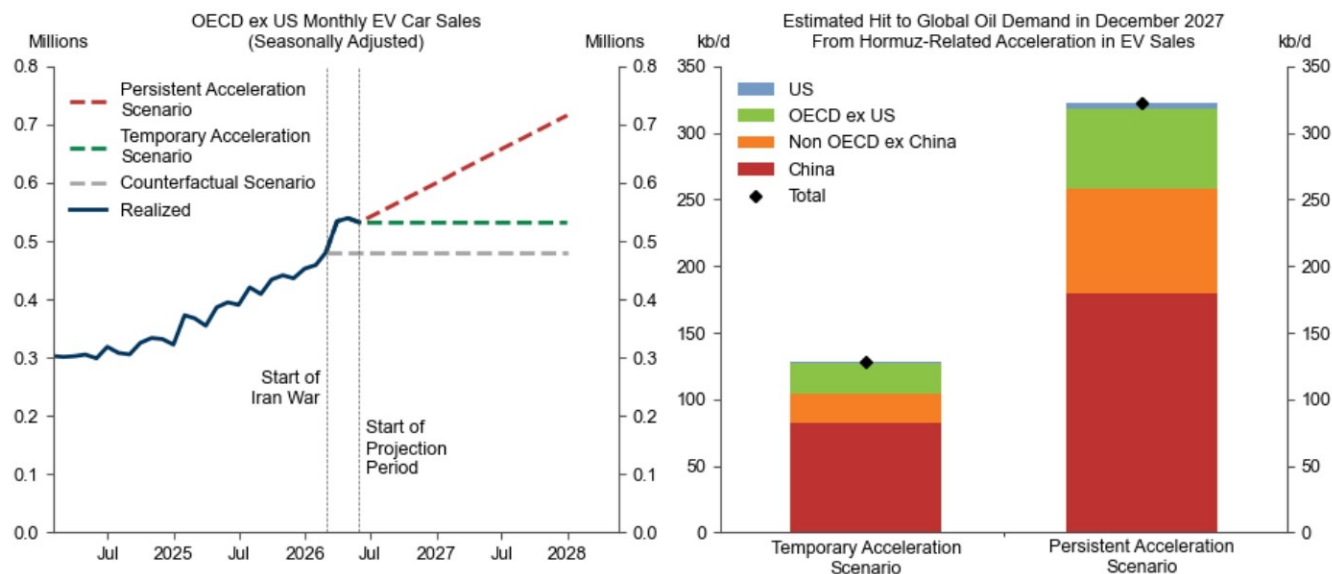
Exhibit 4: 12 Out of the 15 Largest EV Markets Have Experienced an Increase in EV Car Sales Penetration Since February



For 15 largest EV markets by 2025 sales volume. Solid bars indicate that May share is calculated using realized data; striped bars indicate that May EV sales and/or May total car sales data is nowcasted.

South Korea EV car sales surged in February 2026 due to the earlier-than-usual finalization of the federal EV subsidy scheme, which led to rapid purchasing.

Source: Government Data, IEA, Goldman Sachs Global Investment Research

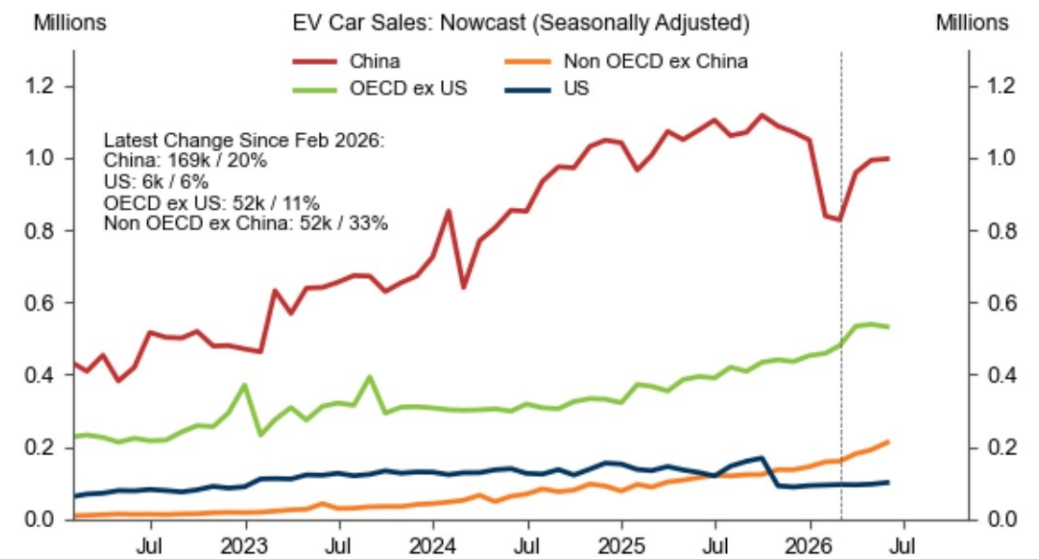
Exhibit 5: We Estimate a Hit to Global Oil Demand in December 2027 of 0.13-0.32mb/d, Depending on the Persistence of the Acceleration in EV Car Sales


The “Temporary Acceleration” scenario assumes that regional EV car penetration rates remain flat at their May 2026 levels. The “Persistent Acceleration” scenario assumes that regional EV car penetration rates grow linearly in accordance with their respective February 2026 – May 2026 trends (through August 2026 before gradually slowing for China, and through December 2027 for world ex China). We assume in both scenarios that total passenger car sales remain flat at February 2026 – May 2026 average levels from June 2026 onwards. The hit to global oil demand is estimated relative to a counterfactual scenario in which regional EV car sales remain flat at their February 2026 levels.

Source: Goldman Sachs Global Investment Research

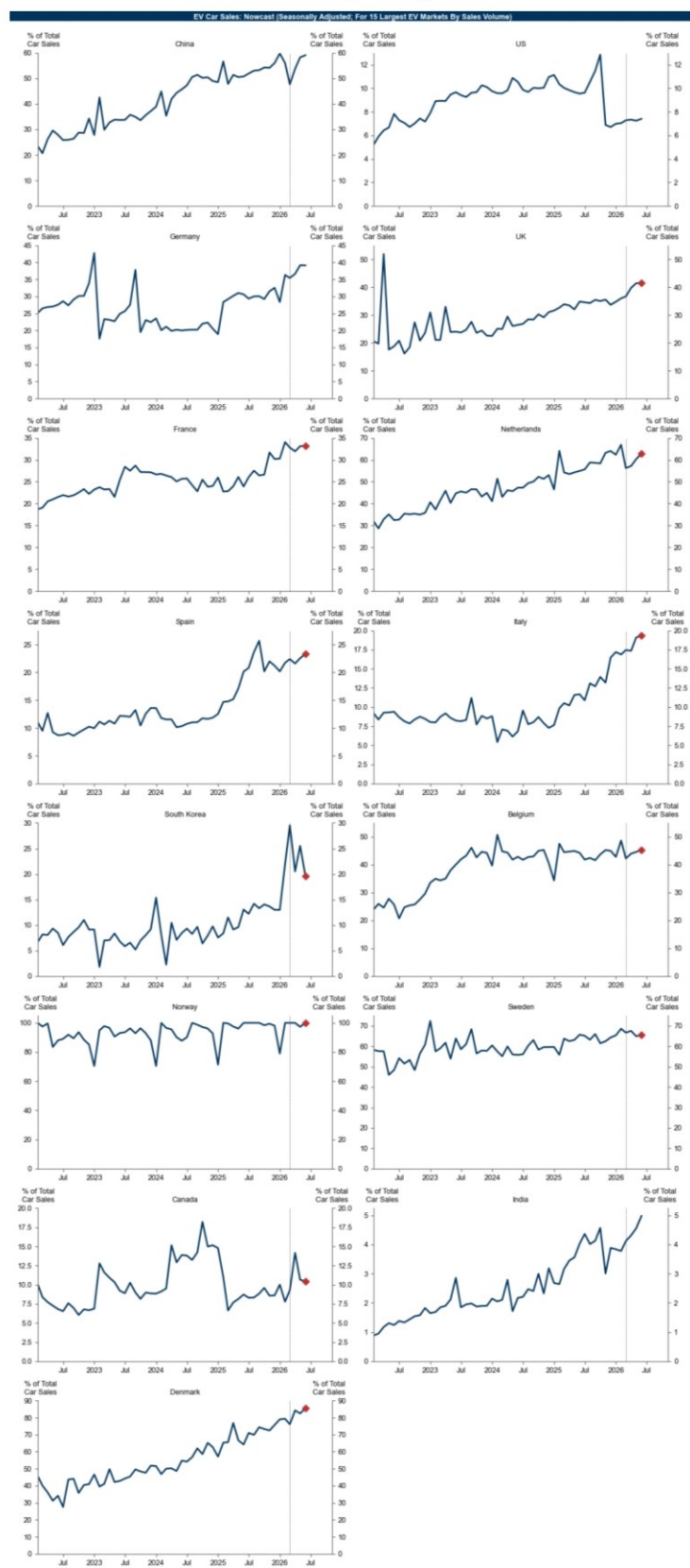
Appendix

Exhibit 6: EV Car Sales Have Accelerated Meaningfully Since February Across the World ex US



Source: Government Data, IEA, Goldman Sachs Global Investment Research

Exhibit 7: The Hormuz-Related Increase in EV Car Sales Penetration Has Generally Been Broad-Based, Although Some Markets Have Experienced Policy Changes



Red diamonds indicate data points that are nowcasted. The vertical line on each chart indicates the start of the Hormuz shock.

Source: Government Data, IEA, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Alexandra Paulus, Daan Struyven, Yulia Zhestkova Grigsby and Filippo Cuscito, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Alexandra Paulus Goldman Sachs & Co. LLC, Daan Struyven Goldman Sachs & Co. LLC, Yulia Zhestkova Grigsby Goldman Sachs & Co. LLC, Filippo Cuscito Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is

defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your

individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.